# PWM con N555 en KiCAD

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Professor/Authority or other information

**Abstract—**Welcome to ...

**Keywords—**...

## 1. Introduction

Welcome to ...

## 2. Análisis

The `\maketitle` command generates the title and author information section, including the professor name and affiliations. The title can be modified in tau-class/tau.cls/title style section.

### 2.1. Objetivo

- ¿Qué hace?
- ¿Cómo es?
- ¿Cuánto cuesta?

### 2.2. Casos de Uso

- ¿Quién lo usa?
- ¿Dónde está instalado?
- ¿Cómo se instala?

### 2.3. Soluciones

- Características
- Requerimientos

## 3. Especificación del Hardware

### 3.1. Diagrama de Bloques

Representación gráfica de los circuitos que lleva la PCB.

### 3.2. Floor Plant

Representación de acomodo de los circuitos dentro de la PCB.

### 3.3. Descripción de Circuitos

Explicación de funcionamiento individual de circuitos.

### 3.4. Pinout

Tabla de conexiones de circuito hacia el microcontrolador.

## 4. Esquemático

1. Creación del Diagrama Electrónico.
2. Agregar características a los símbolos (footprint, número de parte).
3. Generación de netlist.

## 5. PCB Layout

### 5.1. Acomodo mecánico de componentes

### 5.2. Rooteo de la PCB

### 5.3. Validación del DRC

## 6. Acomodo mecánico

### 6.1. Revisión de Serigrafía

### 6.2. Generación del Modelo 3D

## 7. Tables and figures

### 7.1. Tables

Table 1 shows an example table. The `\tabletext{}` is used to add notes to tables easily.

**Table 1.** Astronomical Object Data

| Object | Distance (Light Years) |
|---|---|
| Alpha Centauri | 4.37 |
| Betelgeuse | 642.5 |
| Andromeda Galaxy | 2.537 million |

*Note: The table contains data of some famous celestial objects.*

### 7.2. Figures

Fig. 1 shows an example figure.

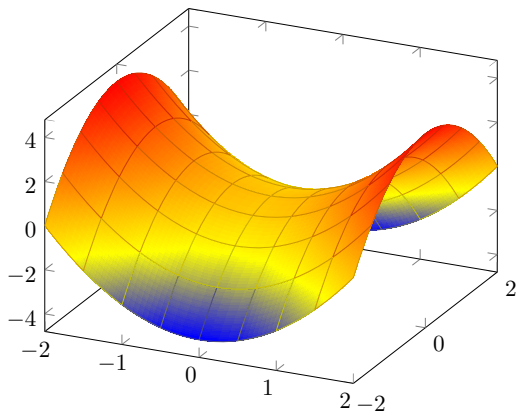



**Figure 1.** Example figure obtained from PGFPlots [1].

Fig. 2 shows an example of two figures that covers the width of the page. It can be placed at the top or bottom of the page. The space between the figures can also be changed using the `\hspace{Xpt}` command.

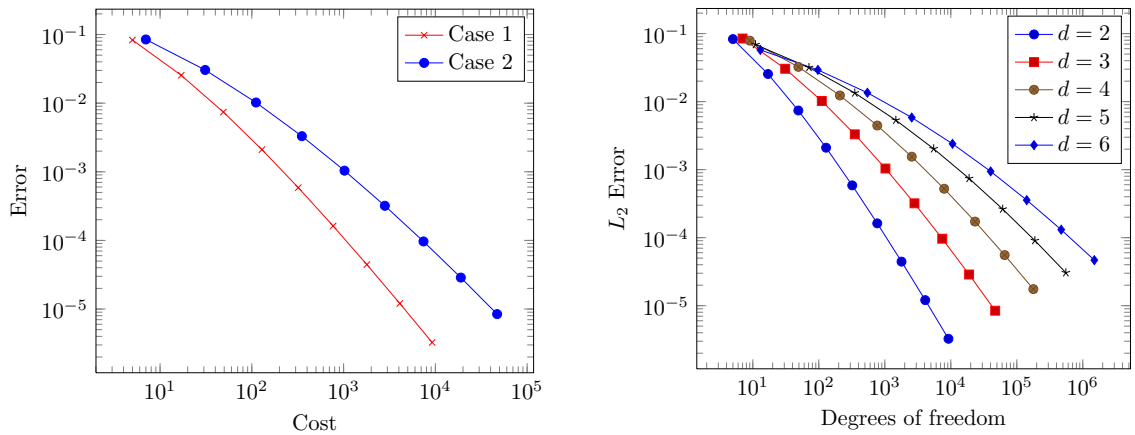
## 8. Tau packages

### 8.1. Tauenvs

This template has its own environment package *tauenvs.sty* designed to enhance the presentation of the document. Among these custom environments are *tauenv*, *info* and *note*.

There are two environments which have a predefined title. These can be included by the command `\begin{note}` and `\begin{info}`. All the environments have the same style.

An example using the tau environment is shown below.




**(a)** Example left figure.

**(b)** Example right figure.

**Figure 2.** Example figure that covers the width of the page obtained from PGFPlots [1].

**Environment with custom title**

This is an example of the custom title environment. To add a title type `[frametitle=Your title]` next to the beginning of the environment (as shown in this example).

Tauenv is the only environment that you can customize its title. On the other hand, info and note adapt their title to Spanish automatically when this language package is defined.

### 8.2. Taubabel

In previous versions, we included a package called *taubabel*, which have all the commands that automatically translate from English to Spanish when this language package is defined.

By default, tau displays its content in English. However, at the beginning of the document you will find a recommendation when writing in Spanish.

*Note:* You may modify this package if you want to use other language than English or Spanish. This will make easier to translate the document without having to modify the class document.

## 9. Equation

Equation 1, shows the Schrödinger equation as an example.

$$\frac{\hbar^2}{2m}\nabla^2\Psi + V(\mathbf{r})\Psi = -i\hbar\frac{\partial\Psi}{\partial t} \tag{1}$$

The *amssymb* package was not necessary to include, because stix2 font incorporates mathematical symbols for writing quality equations. In case you choose another font, uncomment this package in tau-class/tau.cls/math packages.

If you want to change the values that adjust the spacing above and below the equations, play with `\setlength{\eqskip}{8pt}` value until the preferred spacing is set.

## 10. Adding codes

This class[1] includes the *listings* package, which offers customized features for adding codes in LaTeX documents specifically for C, C++, LaTeX and Matlab.

You can customize the format in tau-class/tau.cls/listings style.

```
function fibonacci_sequence(num_terms)
    % Initialize the first two terms of the sequence
    fib_sequence = [0, 1];

    if num_terms < 1
        disp('Number of terms should be greater than or equal to 1.');
        return;
    elseif num_terms == 1
        fprintf('Fibonacci Sequence:\n%d\n', fib_sequence(1));
        return;
    elseif num_terms == 2
        fprintf('Fibonacci Sequence:\n%d\n%d\n', fib_sequence(1), fib_sequence(2));
        return;
    end

    % Calculate and display the Fibonacci sequence
    for i = 3:num_terms
        fib_sequence(i) = fib_sequence(i-1) + fib_sequence(i-2);
    end

    fprintf('Fibonacci Sequence:\n');
    disp(fib_sequence);
end
```

**Code 1.** Example of Matlab code.

If line numbering is defined at the beginning of the document, I recommend placing the command `\nolinenumbers` at the start and `\linenumbers` at the end of the code.

This will temporarily remove line numbering and the code will look better.

## 11. References

The default formatting for references follows the IEEE style. You can modify the style of your references. See appendix for more information.

## 12. Appendix

### 12.1. Alternative title

You can make the following modification in tau-class/tau.cls/title preferences section to change the position of the title.

```
\newcommand{\titlepos}{\centering}
```

**Code 2.** Alternative title.

This will move the title to the center.

### 12.2. Info environment

An example of the info environment declared in the 'tauenvs.sty' package is shown below. Remember that *info* and *note* are the only packages that translate their title (English or Spanish).

[1] Hello there! I am a footnote :)

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**Information**

Small example of info environment.

### 12.3. Equation skip value

With the `\eqskip` command you can change the spacing for equations. The default *eqskip* value is 8pt.

```
\newlength{\eqskip}\setlength{\eqskip}{8pt}
  \expandafter\def\expandafter\normalsize\expandafter{%
    \normalsize%
    \setlength\abovedisplayskip{\eqskip}%
    \setlength\belowdisplayskip{\eqskip}%
    \setlength\abovedisplayshortskip{\eqskip-\
     baselineskip}%
    \setlength\belowdisplayshortskip{\eqskip}%
  }
```

**Code 3.** Equation skip code.

### 12.4. References

In case you require another reference style, you can go to tau-class/tau.cls/biblatex and modify the following.

```
\RequirePackage[
  backend=biber,
  style=ieee,
  sorting=ynt
]{biblatex}
```

**Code 4.** References style.

By default, *tau class* has its own .bib for this example, if you want to name your own bib file, change the *addbibresource*.

```
\addbibresource{tau.bib}
```

## 13. FAQ

### How do I manage my references?

To manage your references, I recommend using the tool scribbr. You can simply enter the URL or create your own citation, and then export it to LaTeX using the options in the three-dot menu.

The generated citation can be copied and pasted into *tau.bib*, the file designated for bibliography management. You may rename this file, but if you do, remember to update the `\addbibresource` command in *tau.cls* under the *biblatex* section.

**Note**

Some platforms, such as Google Scholar or scientific journals, provide citations directly in LaTeX format. Therefore, check if there is a "how to cite this document" section to streamline the citation process even further.

Here is an example of a reference code compatible with BibTeX.

```
@misc{PFGPlots,
    author = {PFGPlots},
    title = {A LaTeX package to create plots.},
    url = {https://pgfplots.sourceforge.net/}
}
```

If you have any further questions, you can refer to the following page.

- Bibliography management with biblatex

An example of an IEEE-formatted bibliography can be found in Fig. 1. If you are using a different citation style, such as APA, I recommend using the `\parencite` command to automatically include parentheses around citations.

### What should I do with the example files?

If you edit this template, you can remove the example figures, the bibliography entries in *tau.bib*, and the *fibonacci.m* Matlab code without affecting your document.

### How do I convert my document into a column?

At the beginning of the document, you will find a `\documentclass` command. By default, it will show *twocolumn*. Simply change this to *onecolumn* and recompile the document.

If further adjustments are needed, you will have to go to *tau.cls* and navigate to the relevant section (as *geometry package*) to make any other changes or modifications.

### How do I change the paper size?

Similarly, you can change the paper size (by default, this class was adapted for A4 size). The following paper sizes are available in LaTeX:

- letterpaper ($11 \times 8.5$ in)
- legalpaper ($14 \times 8.5$ in)
- executivepaper ($10.5 \times 7.25$ in)
- a4paper ($21 \times 29.7$ cm)
- a5paper ($21 \times 14.8$ cm)
- b5paper ($25 \times 17.6$ cm)

### How do I place equations easily?

For equations, we have two options: inline or on its own line. For inline equations, simply place a dollar sign ($) at the beginning and end of the equation. However, if you want the equation to be displayed on its own line, you need to use the equation environment.

If you find it challenging to write formulas directly in LaTeX, you can use text editors like Word. In the equations menu, you can select LaTeX in the conversion section and copy and paste the equation you wrote into one of these two environments.

## 14. Contact me

You can contact me through these methods.

WiX https://memonotess1.wixsite.com/memonotess
memo.notess1@gmail.com
memo.notess

## 15. Supporting

Did you like this class document? Check out our new project the rho class, made for complex articles and reports.

### Any contributions are welcome!

Coffee keeps me awake and helps me create better LaTeX templates. If you wish to support my work, you can do so through PayPal: **https://www.paypal.me/GuillermoJimeenez**.

Enjoy writing with tau LaTeX class

## References

[1] PFGPlots, *A latex package to create plots.* [Online]. Available: https://pgfplots.sourceforge.net/


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